

## Problem

Your laboratory has available a large number of  $5\text{-}\mu\text{F}$  capacitors rated at  $150\text{ V}$ . To design a capacitor bank of  $10\text{ }\mu\text{F}$  rated at  $600\text{ V}$ , how many  $5\text{-}\mu\text{F}$  capacitors are needed and how would you connect them?

## Step-by-step solution

## Step 1 of 2

Use  $5\text{ }\mu\text{F}$  capacitors rated at  $150\text{ V}$  to design a capacitor bank of  $10\text{ }\mu\text{F}$  rated at  $600\text{ V}$ .

The number of capacitors required for each block to achieve the rated voltage is,

$$\begin{aligned} n &= \frac{600}{150} \\ &= 4 \end{aligned}$$

The summation of voltage values is possible in series configuration. Hence, four  $5\text{ }\mu\text{F}$  capacitors are connected in series manner.

The resultant capacitance of 4 series connected capacitors is,

$$\begin{aligned} C &= \frac{\left[ \frac{(5)(5)}{5+5} \right] \left[ \frac{(5)(5)}{5+5} \right]}{\left[ \frac{(5)(5)}{5+5} \right] + \left[ \frac{(5)(5)}{5+5} \right]} \\ &= \frac{(2.5)(2.5)}{2.5+2.5} \\ &= 1.25\text{ }\mu\text{F} \end{aligned}$$

## Step 2 of 2

The required value of capacitance is  $10\text{ }\mu\text{F}$ . Therefore, it is required to connect the capacitor blocks in parallel, because the voltage across two points is same in parallel configuration.

The number of capacitors blocks required to achieve the required capacitance is,

$$\begin{aligned} n &= \frac{10 \times 10^{-6}}{1.25 \times 10^{-6}} \\ &= 8 \end{aligned}$$

Hence, total of eight groups in parallel with each group made up of four capacitors in series to design a capacitor bank of  $10\text{ }\mu\text{F}$  rated at  $600\text{ V}$ .